Yichen Andy Yu

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Education

North Carolina State University — *Raleigh, USA*
PhD in Computer Science — *Aug 2026 – Present*

Master of Science in Computer Science — *Aug 2025 – Present*

University of Rochester — *Rochester, USA*
Master of Science in Computer Science — *Aug 2023 – May 2025*

- Advisor: *Prof. Zhen Bai and Prof. Dillon Dzikowicz*

University of Wisconsin–Madison — *Madison, USA*
VISP Student in Computer Science — *Jan 2023 – Jun 2023*

Feng Chia University — *Taichung, Taiwan*
Bachelor of Science in Computer Science — *Sep 2020 – Jun 2023*

- Advisor: *Prof. Ming-Yen Lin*

Research Experience

North Carolina State University — *Raleigh, USA*
Research Staff (Mentor: *Prof. Qiao Jin*) — *Aug 2025 - Present*

- Conducting VR safety research, leveraging AI for safety classification, and developing highly immersive replacement solutions. (Advisor: *Dr. Qiao Jin*)
- Prototyped and fabricated custom 3D-printed mounts and accessories to integrate sensing hardware with VR headsets, enabling robust, repeatable experimental setups.

Carnegie Mellon University — *Pittsburgh, USA*
Research Assistant — *Nov 2024 – Aug 2025*

University of Rochester Medical Center - Dzikowicz Lab — *Rochester, USA*
Research Assistant — *Mar 2024 – Present*

- Digitized paper-based ECG charts into structured, machine-readable data formats for integration with clinical datasets and downstream physiological analysis.
- Analyzed wearable ECG signals recorded during physical activities (e.g., 6MWT and veloergometry) to assess cardiac responses in post-operative rehabilitation patients.
- Extracted heart rate dynamics, HRV features, and age-predicted max HR to evaluate exercise intensity, cardiac effort, and post-activity recovery trends.

ROC-HCI Group — *Rochester, USA*
Research Assistant — *Aug 2023 – Aug 2025*

- Used Unity to create AR educational games for K-12 kids on Android and Meta Quest 3.
- Used Unity to develop an MR system to alleviate separation anxiety disorder in kids.
- Designed an AR-based system for parents of kids with ASL to learn sign language with their kids.

Publications

Yichen Andy Yu, Wanru Li, Qiaoran Wang, Jymon Ross, Gavin Johnson, Mandy Lui, Qiao Jin. SpatialPrompt: XR-Based Spatial Intent Expression as Executable Constraints for AI Generative 3D Design. (DIS Poster'26)

Qiao Jin, Conrad Borchers, Ashish Gurung, Sean Jackson, Sameeksha Agarwal, Jocelyn Wang, **Yichen Yu**, Pragati Maheshwary, Vincent Aleven. Sticky Help, Fleeting Effects: Session-by-Session Analytics of Teacher Interventions in K–12 Classrooms. (LAK'25)

Yichen Yu*, Huan-Song Xu*. RunPacer: A Smartwatch-Based Vibrotactile Feedback System for Symmetric

Co-Running by Visually Impaired Individuals and Guides (ASSETS Poster'25). https://doi.org/10.1145/3663547.3759738

Yichen Yu*, Yifan Jiang*, Mandy Lui, Qiao Jin. GenLARP: Enabling Immersive Live Action Role-Play through LLM-Generated Worlds and Characters (ISMAR Poster'25). https://10.1109/ISMAR-Adjunct68609.2025.00178

Xiaofei Zhou*, Yunfan Gong*, **Yichen Yu**, Zhenyao Cai, Zhen Bai. Designing Embodied Metaphors and Analogies for AI Literacy: An Iterative Study with Experts, Teachers, and Children. (ACM TOCE). https://dl.acm.org/doi/10.1145/3773901

Qiaoran Wang*, **Yichen Yu***. Noetic Dream: A Personalized VR and Meditation System for Lucid Dream Training (UIST Poster'25). https://doi.org/10.1145/3746058.3758424

Yichen Yu*, Qiaoran Wang*. NieNie: Real-Time Stress Detection and Interactive Squeezing Rhythm through Unity Gameplay and Language Model Guidance (Ubicomp SC'25). https://doi.org/10.1145/3714394.3750586

Yichen Yu, Qiao Jin. Chameleon: Unobtrusive Substitution of Real-World Obstacles in VR with Risk-Level-Aware Adaptation. (CHI LBW'25). https://doi.org/10.1145/3706599.3719779

Xiaofei Zhou*, Yunfan Gong*, **Yichen Yu**, Yi Zhang, Jeremy Smith, Zhen Bai. Design AI for My Community: A Case Study in a Freedom-to-Read Summer Camp. (ISLS'25). https://doi.org/10.22318/icls2025.111198

Alexander Bae, **Yichen Yu**, Chi-Ju Lai, Wendy Brunner, Nicole Krupa, Mary Carey, Wai Cheong Tam, Dillon Dzikowicz. Structural Heart Abnormalities are Prevalent on the 12-lead ECG among Firefighters. (Journal of Occupational and Environmental Medicine). https://10.1097/JOM.0000000000003409

Alexander Bae, **Yichen Yu**, Chi-Ju Lai, Wendy Brunner, Nicole Krupa, Mary Carey, Wai Cheong Tam, Dillon Dzikowicz. 9-Year Longitudinal Assessment of the 12-lead Electrocardiogram of Volunteer Firefighters. (American Heart Association Scientific Sessions'24). https://doi.org/10.1161/circ.150.suppl$_1$.4137975

Academic Activities

Reviewer Experience: ACM C&C'25, ACM CHI' 25, ACM IUI' 26

Teaching Experience

CSC216 & CSC416 - AR/VR Design — *Rochester, USA*
Teaching Assistant — *Aug 2024 – Dec 2024*

Milele Chikasa Anana Elementary School — *Madison, USA*
Teacher — *Jan 2023 – Jun 2023*

- Taught K-12 elementary school students basic programming skills in the Scratch Club.

Work Experience

Porsche Engineering — *Shanghai, China*
HMI System Engineer Intern — *May 2024 – Aug 2024*

- Conducting VR safety research, leveraging AI for safety classification, and developing highly immersive replacement solutions. (Advisor: *Dr. Qiao Jin*)
- Prototyped and fabricated custom 3D-printed mounts and accessories to integrate sensing hardware with VR headsets, enabling robust, repeatable experimental setups.

Competition Projects

Running Training Assistance App for the Blind - RunPacer
This project has been acquired for about 33,000 USD (1,000,000 NTD).
1st Prize in 2023 Apple IOS Mobile Application Innovation Competition, invited to attend WWDC 2023.

- Helped blind users connect with running partners online, while used WatchKit, HealthKit, and ClockKit to track and display key health metrics during their runs.
- Collected hydration and health data, offering personalized real-time reminders that contributed to a 30% decrease in potential dehydration incidents. SwiftUICharts analyzed performance metrics post-run, enhancing user experience by tracking progress visually in Summary View.
- Analyzed the collected data using SwiftUICharts after each session and presented the results in charts through Summary View, helping users track performance and health metrics.

Research Mentoring

Aaron Joseph	*Jun 2026-Now*
Jymon Ross	*Aug 2025-Now*
Gavin Johnson	*Aug 2025-Now*
Mandy Lui	*May 2025-May 2026*
Yifan Jiang	*May 2025- Jan 2026*
Kyle Jhong	*May 2024- January 2025*
Nicholas DeJesse	*May 2024- January 2025*